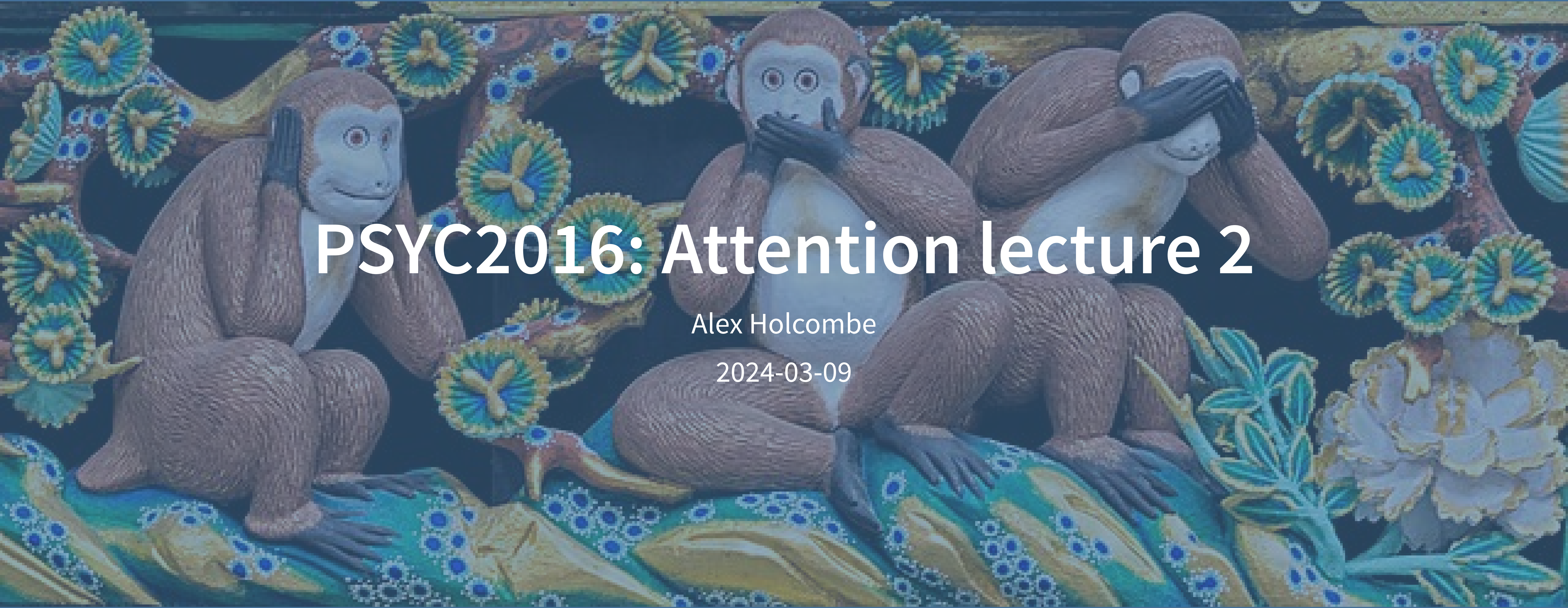


PSYC2016: Attention lecture 2

Alex Holcombe

2024-03-09

Bottleneck

What are you unclear on from today's lecture?

Bottleneck	bottleneck problem	bottlenecks
the bottleneck thing	what actually is a bottleneck	bottlenecking???
bottlenecks. what is a bottleneck? mental? physical? no definition	what is a bottleneck? Is it just a gap between perception and processing?	Processing before bottleneck
Are physical and mental bottlenecks referring to the same thing or are they 2 different concepts	the idea of the central bottleneck did not feel properly defined as a concept, more just felt like a buzz word to refer to any sort of cognition	so researchers don't know why there is a bottle neck effect then?

I've updated the textbook with an additional paragraph on this.

Bottleneck

- Imagine a motorway with many lanes. Everyone has to pay a toll.
- But the government doesn’t have enough money to pay for more than one toll booth (this is in a country that doesn’t have e-tags to collect tolls).
- The capacity to process tolls is equal to one car at a time.
- Thus everyone has to slow down when they near the toll booth because only one car can be processed at a time.

Overt and Covert Attention

6 responses

- Covert and overt attention
- overt vs covert attention
- Still about overt VS covert
- did we start attention already??
- sometimes the unconscious functions are made aware (for example im aware of my breathing), suddenly you cant breath normally anymore, but need to "forget" about it so it could return to normal. why?
- obligatory caleb owens mention

Read Chapter 4 and then give me a more specific question.

What are you unclear on from today's lecture?

Two-Letter Task

4 responses

- What the two-letter experiment proves
- why it is possible for us to read at high speed (think speed reading tests) but can only process one letter at a time
- Whether the two letter task is actually a good paradigm for attention, or whether it's just a visual memory thing, or how those relate
- the letter thing

Suggests limited processing capacity. For another task, see Ch. 3.4

link

Mentimeter survey

most-focused experience

shutting out the world

TODAY

- Chapters 1-4: Bottlenecks, Overt and covert attention
- Vigilance
- Top-down vs. bottom-up attention (Ch. 5)
- Location selection (Ch. 7)
- Feature selection (Ch. 7)
- Absence of combined selection (Ch. 7)

VIGILANCE

- For ecologists, *vigilance* is monitoring surroundings for predators
- Hard to get close to wallabies. Why? Vigilance.
- The arrival of predators is an example of something that can be more important than whatever an animal is doing

- 
- Because of predators many animals (including most mammals!) are nocturnal.

Nocturnal ancestors

- Prior to the dinosaurs, our ancestors had vision specialized for daytime.
- During the dinosaurs, our ancestors shifted into the night
- 

Our mind evolved in an environment of predators

- Vigilance - periodically “look” around
- Balance concentration against watching out for more important things appearing

Vigilance in deer, and monkeys

- player configuration error

  Error 153

  Error 153

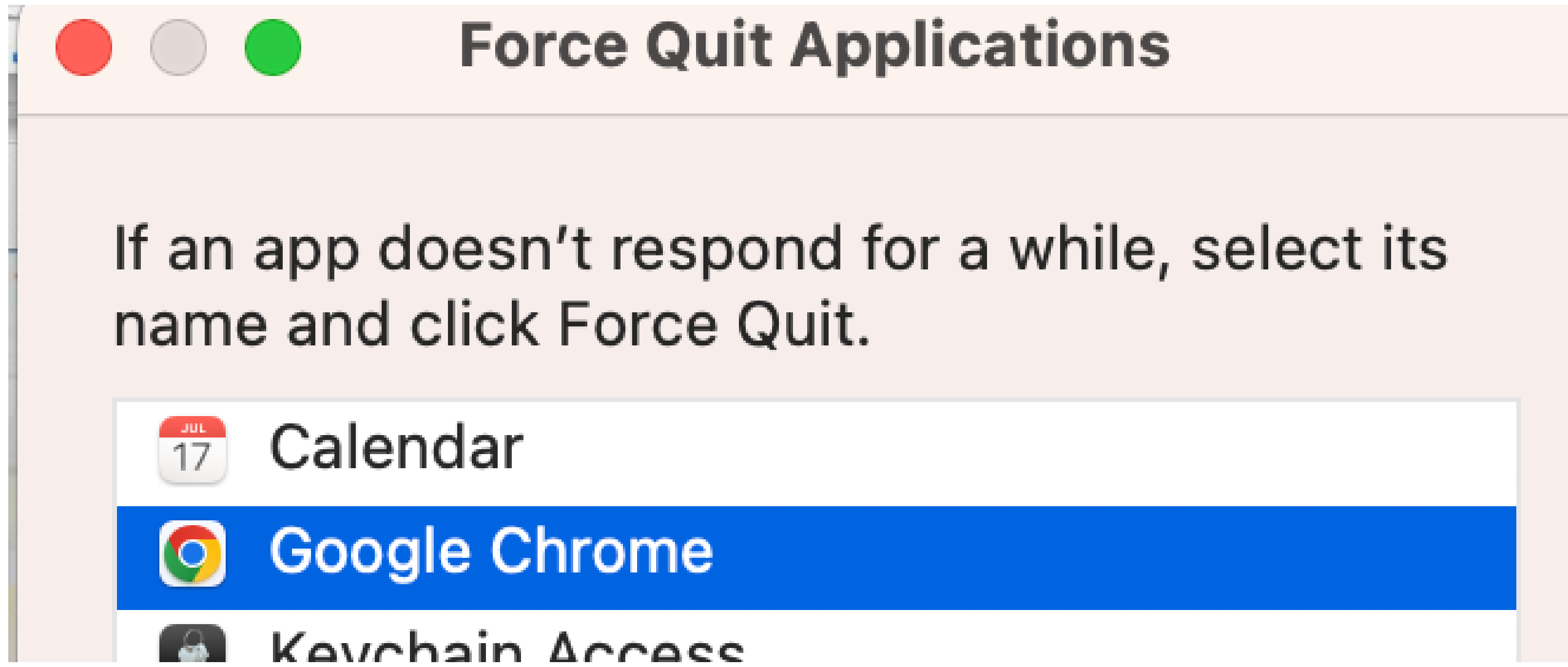
Concentration vs. vigilance

- Solution 1: Alternate between concentrating and looking around
- Solution 2: Have a system for overcoming concentration with potentially-important signals

Computers were not designed with vigilance in mind

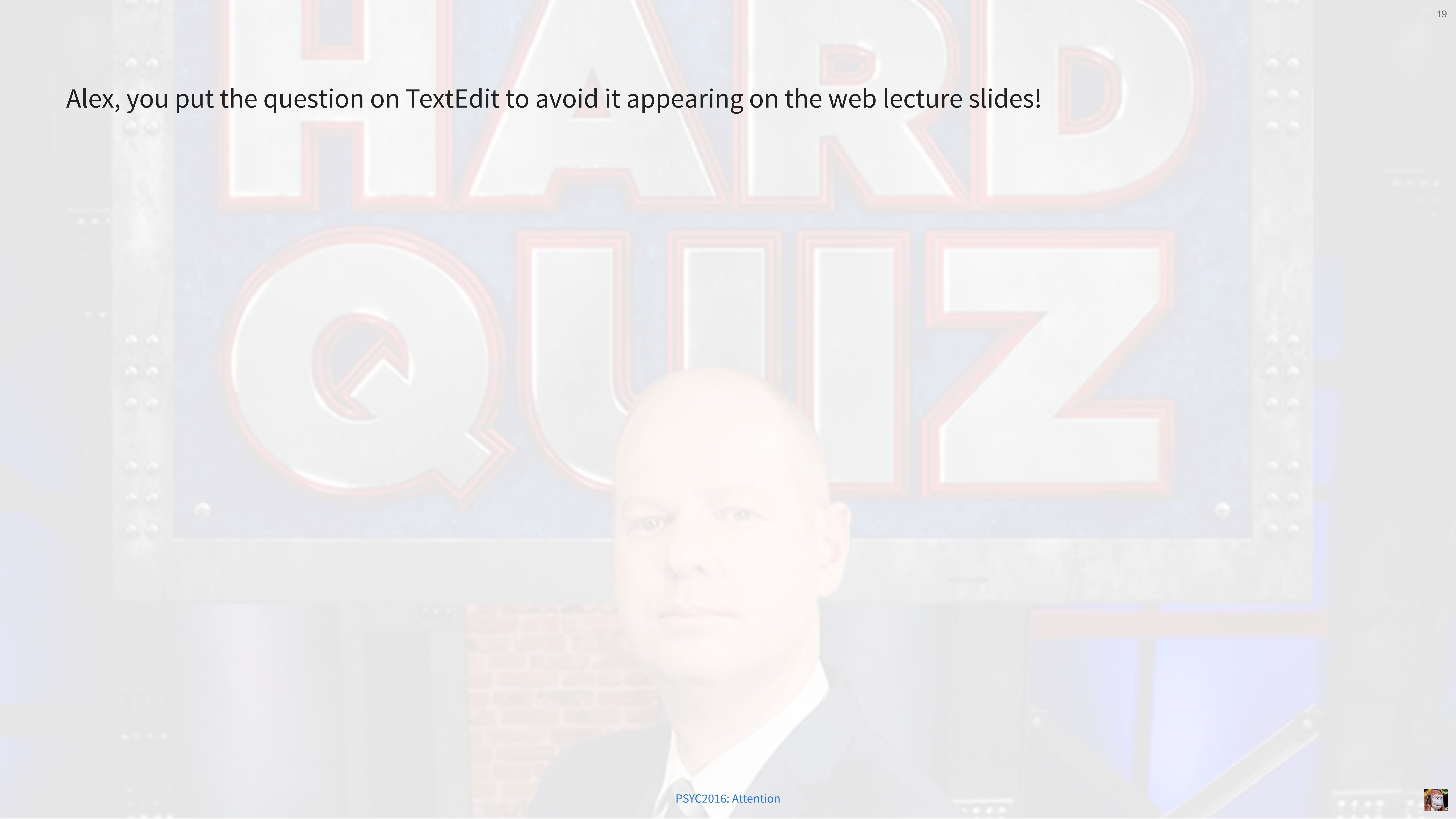
Computer scientists created “interrupt signals”

- Keystroke interrupt signal
  - Option-Command-ESC (MacOS); Alt-F4 (Windows)

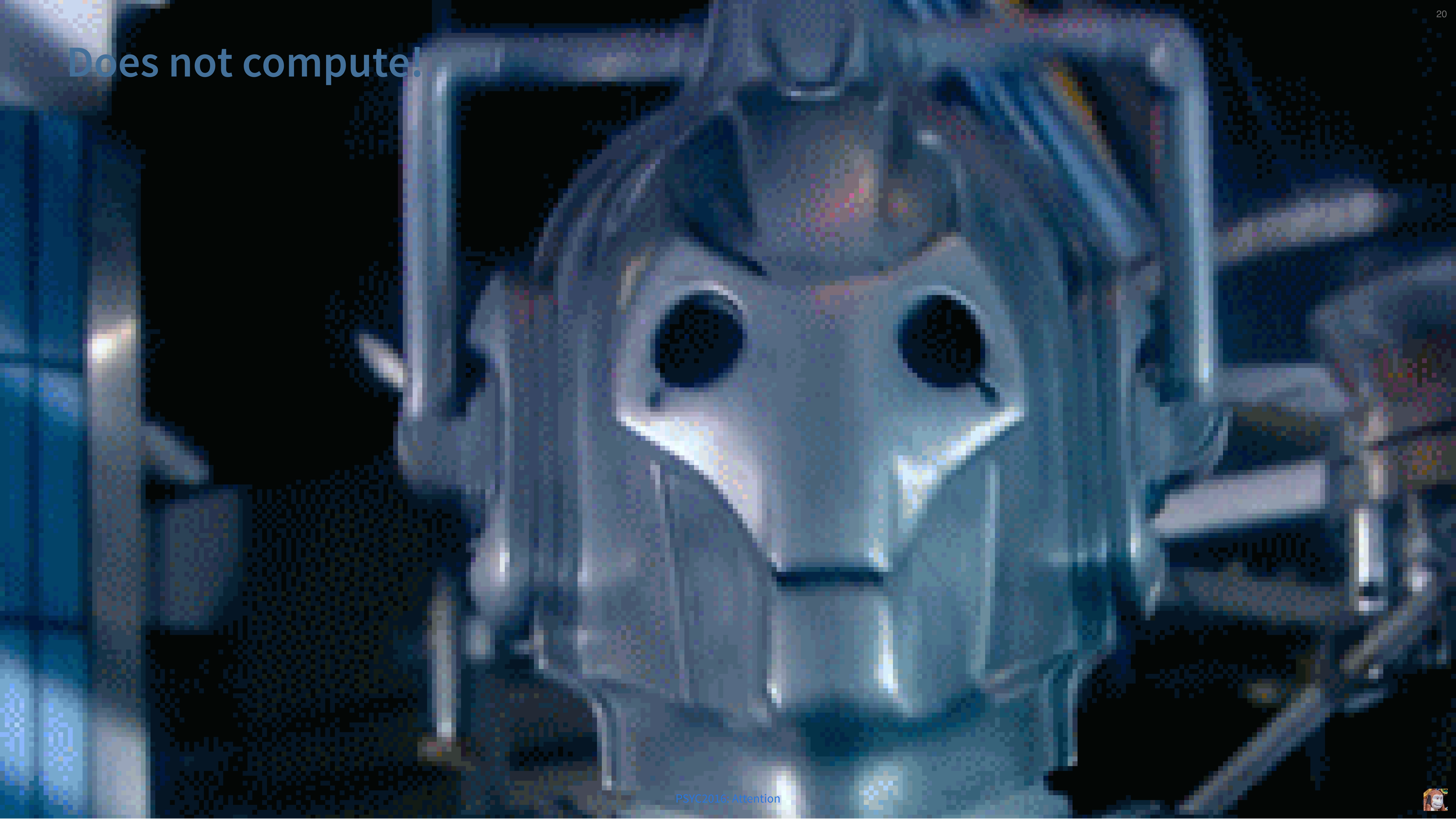



HARD
QUIZ

Alex, you put the question on TextEdit to avoid it appearing on the web lecture slides!



Does not compute!



BOTTOM-UP ATTENTION

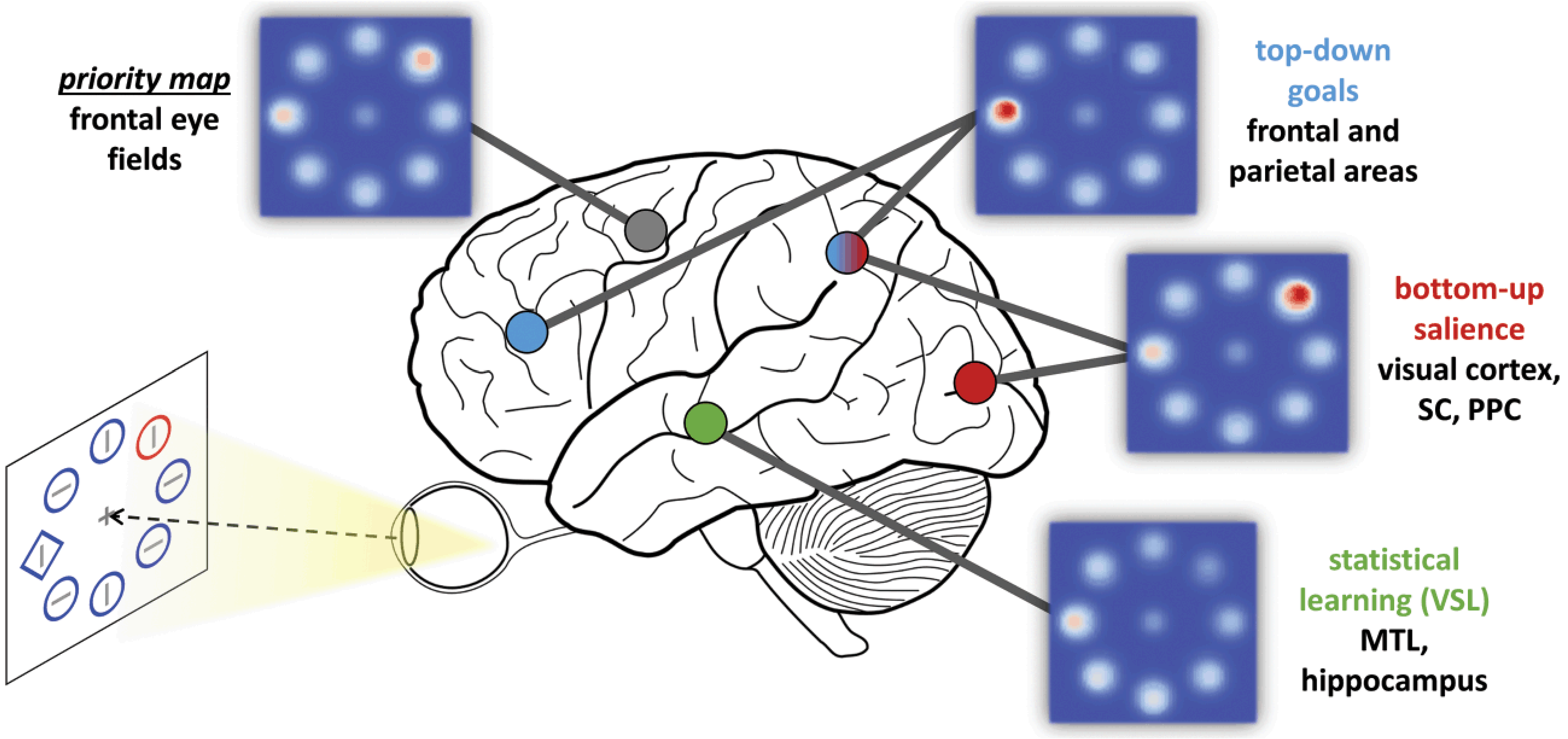
Bottom-up attention

- “Salient” stimuli
- What is salient to us?
  - Different than other stimuli in terms of low-level features, see Chapters 7 and 9
  - Intense stimuli
- Is a kind of vigilance
  - Basic alerting system
  - Like a smoke alarm: dumb, but requires few resources. Cheap to run it all the time.

Top-down attention

- Sometimes called *endogenous* attention
- Internally driven
- Goal directed

Top-down combines with bottom-up (Ch. 5)



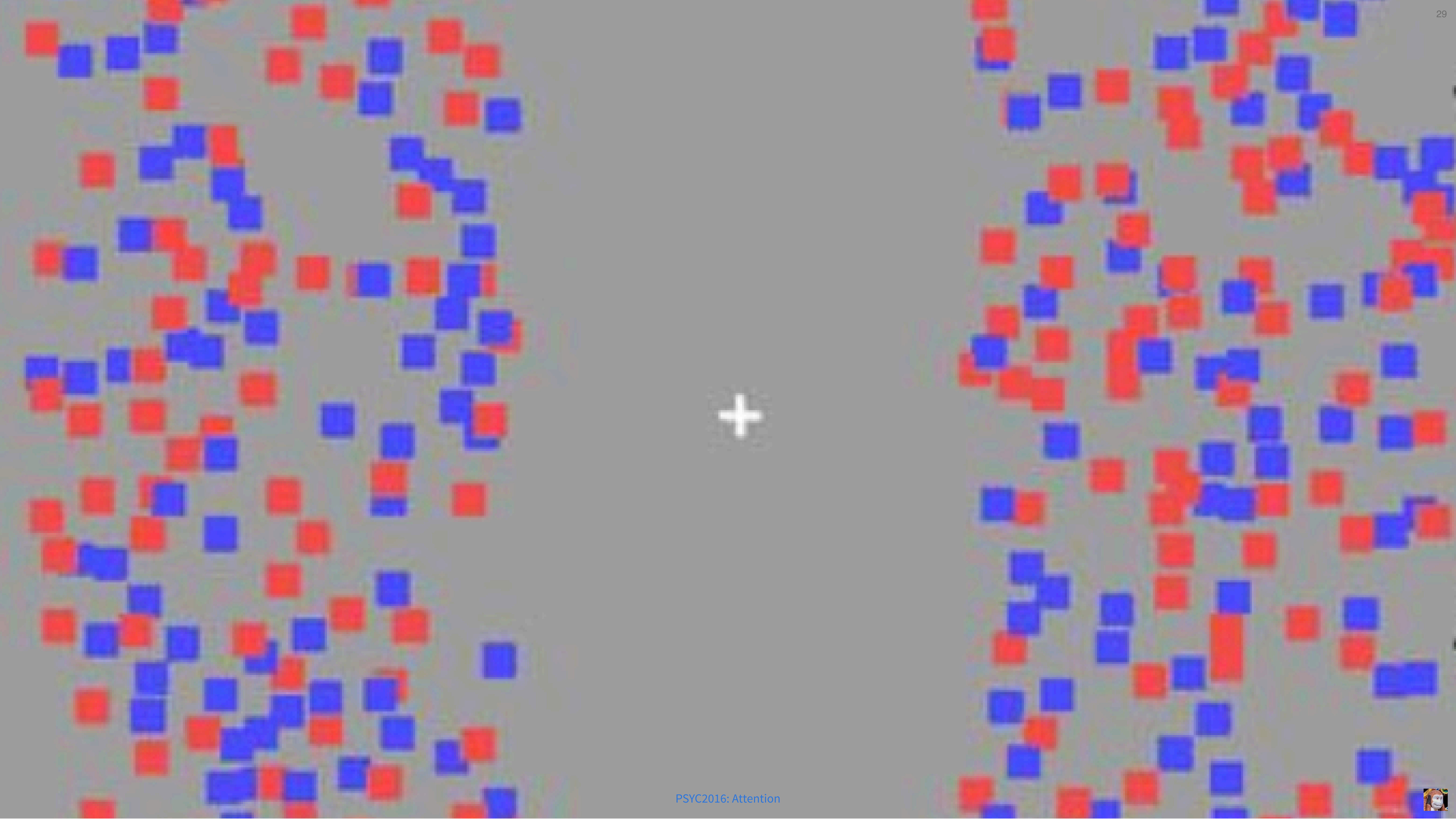
Ways to select a stimulus

- Spatial/object selection
- Feature selection

Combinations of features can’t be directly selected (Ch. 7)

- Features are processed somewhat separately by the brain.
- Conjoining them requires a capacity-limited process

Feature selection is compulsorily global (Ch. 7)



Feature selection is compulsorily global (Ch. 7)

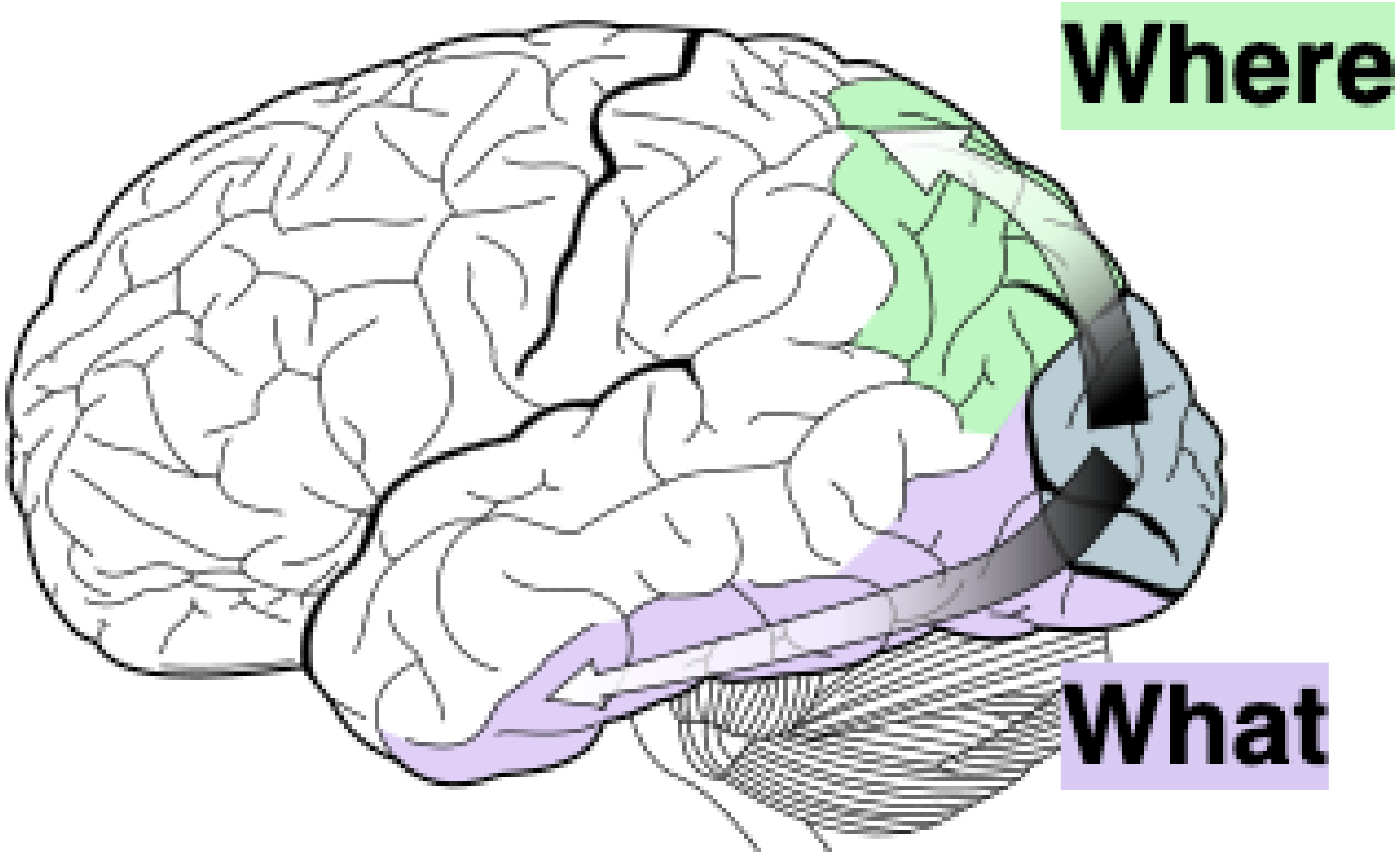
Can’t attend to red on left side while simultaneously attending to blue on right side.

Feature selection is compulsorily global

The *combination* of location and feature selection can’t be done.

You can either select a simple feature everywhere (e.g., red objects), or select the stimuli in a particular region, but you can’t combine those two abilities.

Global: Combining location and feature selection can’t be done

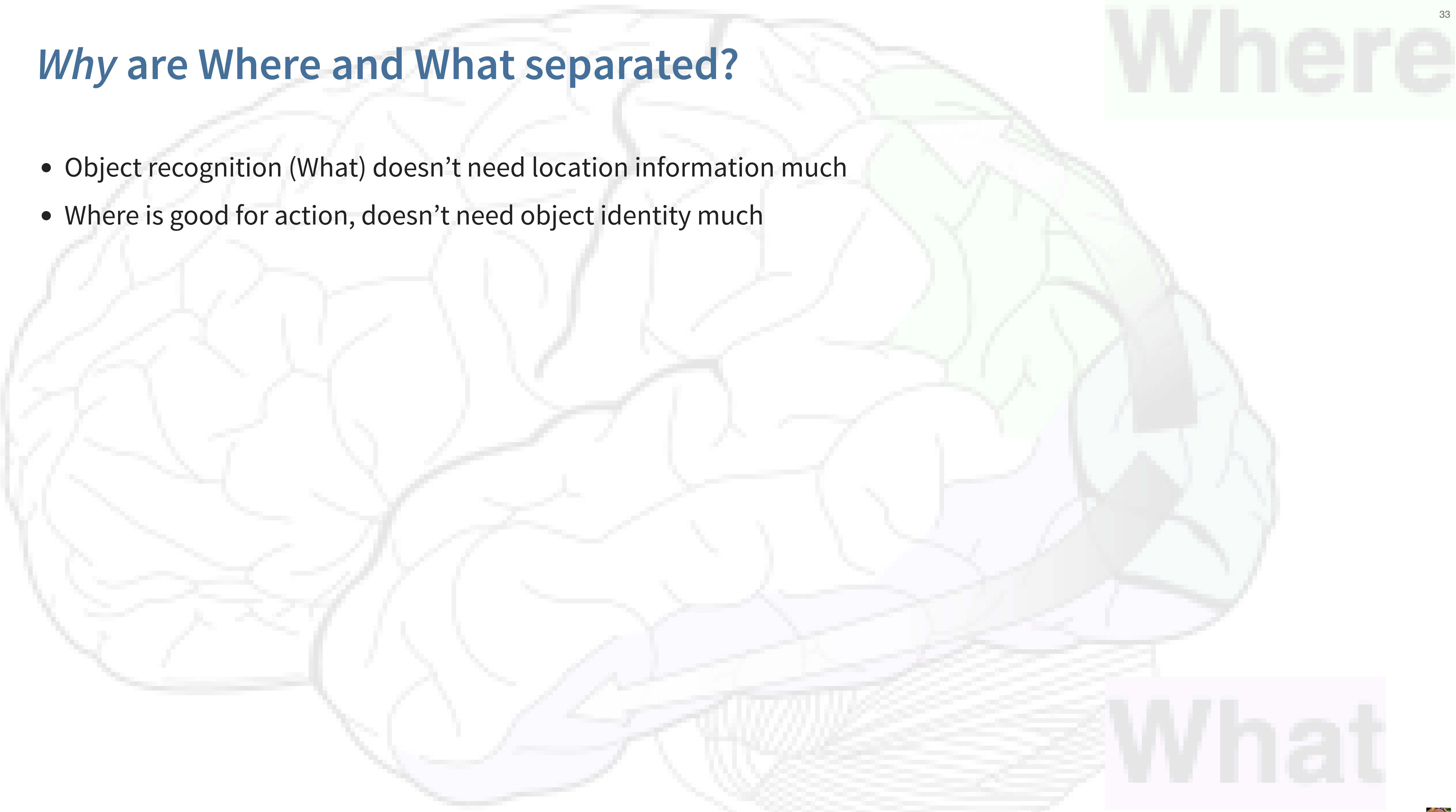


Separation of what and where pathways.

- dorsal = parietal = where = location
- ventral = temporal = what = object recognition

Why are Where and What separated?

- Object recognition (What) doesn't need location information much
- Where is good for action, doesn't need object identity much



TODAY

- Top-down vs. bottom-up attention
- Location selection
- Feature selection
- Absence of combined selection

Thing(s) I didn't understand

Waiting for responses ...